

AI-POWERED HOSPITAL MANAGEMENT SYSTEM

Workflow Documentation

Submitted By:

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Technologies Used

- n8n Workflow Automation
- Google Gemini AI
- Supabase
- Gmail API
- JavaScript

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1. Introduction

Healthcare organizations continue to rely on numerous manual administrative processes for patient registration, appointment scheduling, report management, billing, and patient follow-up. These traditional approaches increase the workload of healthcare staff, consume valuable time, and often lead to delays, inconsistencies, and communication gaps.

The **AI-Powered Hospital Management System** is designed to automate these repetitive processes using **n8n workflow automation**, **Google Gemini AI**, **Supabase**, and **Gmail API**. Instead of replacing healthcare professionals, the system acts as an intelligent assistant that supports doctors by providing preliminary AI-generated recommendations while ensuring that the final clinical decision remains under human supervision.

The solution integrates multiple independent workflows into one automated ecosystem where patient information flows seamlessly between registration, appointments, AI analysis, doctor approval, medication reminders, and billing.

2. Project Overview

The project consists of six interconnected workflows that automate different stages of the patient care lifecycle.

These workflows include:

- Hospital Patient Registration
- Doctor Appointment Booking
- Medical Report Upload & AI Analysis
- AI Doctor Approval System
- Medication & Follow-up Reminder
- Billing & Hospital Analytics

Each workflow performs a specific task while exchanging data with the others through a centralized Supabase database.

The overall objective is to reduce manual work, improve efficiency, minimize errors, and provide faster healthcare services while maintaining complete transparency and human oversight.

3. Technology Stack

Technology	Purpose
n8n	Workflow Automation Platform
Google Gemini AI	AI-powered medical assessment
Supabase	Cloud Database
Gmail API	Email Notifications
JavaScript	Business Logic & ID Generation
Forms	Patient Data Collection

4. Workflow 1 – Hospital Patient Registration

Objective

The Patient Registration Workflow automates the process of registering new patients into the hospital management system. Instead of manually entering patient information into registers or spreadsheets, hospital staff or patients can submit their details using a digital form.

The workflow validates the information, generates a unique Patient ID, stores the patient record in the Supabase database, and sends a confirmation email.

Trigger

The workflow begins when a patient submits the online registration form.

Inputs

- Full Name
- Age
- Gender
- Phone Number
- Email Address
- Blood Group
- Address
- Emergency Contact
- Medical History
- Allergies
- Current Medications

Workflow Process

Step 1

The patient completes the registration form.

↓

Step 2

The workflow checks whether the patient already exists in the database using the email address or phone number.

↓

Step 3

If a duplicate patient is found, registration is prevented.

↓

Step 4

If no duplicate exists, JavaScript generates a unique Patient ID.

↓

Step 5

The patient information is stored in the Supabase database.

↓

Step 6

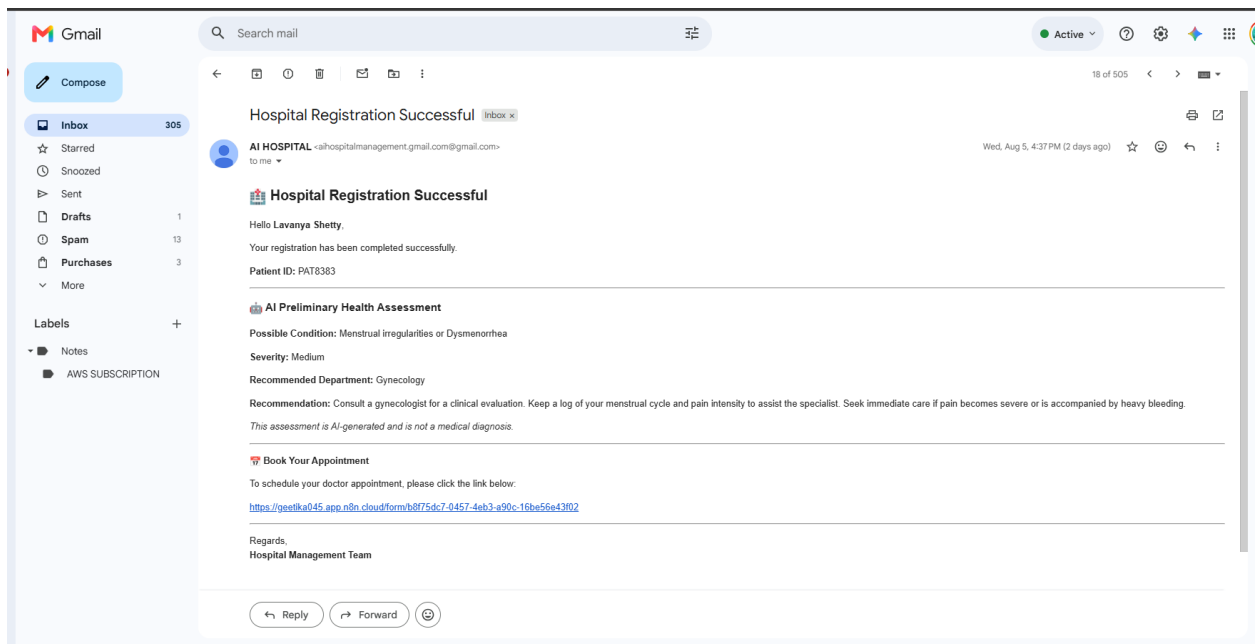
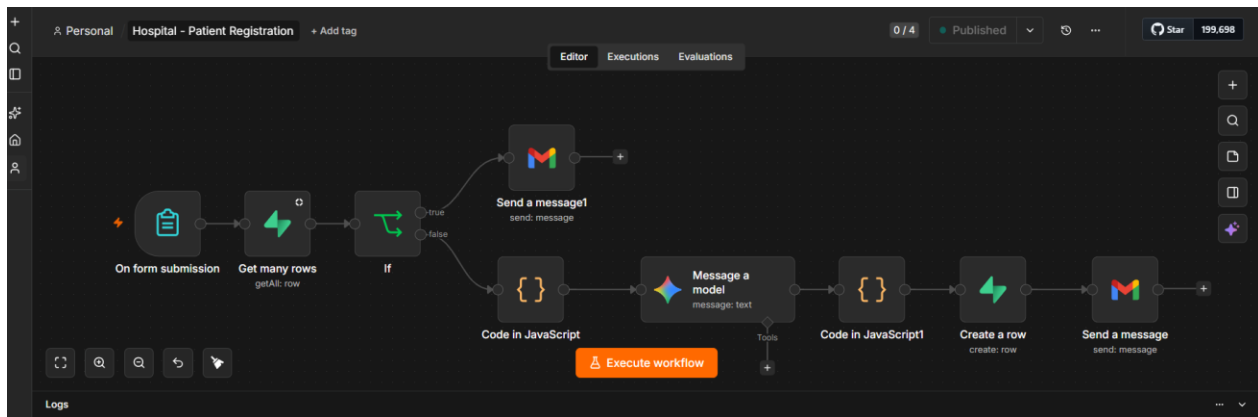
A confirmation email is automatically sent to the registered email address.

Output

- Unique Patient ID
- Patient Record Stored
- Confirmation Email

Benefits

- Eliminates duplicate registrations
- Reduces paperwork
- Creates centralized digital records
- Improves registration speed



5. Workflow 2 – Doctor Appointment Booking

Objective

The Appointment Booking Workflow allows patients to request appointments while using AI to determine the most appropriate medical department based on the symptoms entered.

The workflow automatically checks patient details, predicts the required department using Gemini AI, finds an available doctor, books the appointment, stores the appointment information, and sends an email confirmation.

Trigger

Appointment booking form submission.

Inputs

- Patient ID
- Symptoms
- Preferred Appointment Date
- Preferred Time

Workflow Process

Step 1

Patient submits appointment request.

↓

Step 2

System validates Patient ID.

↓

Step 3

Patient details are retrieved from Supabase.

↓

Step 4

Gemini AI analyzes symptoms.

↓

Step 5

AI predicts the most appropriate department.

↓

Step 6

Available doctor is retrieved.

↓

Step 7

Appointment ID is generated.

↓

Step 8

Appointment is stored in Supabase.

↓

Step 9

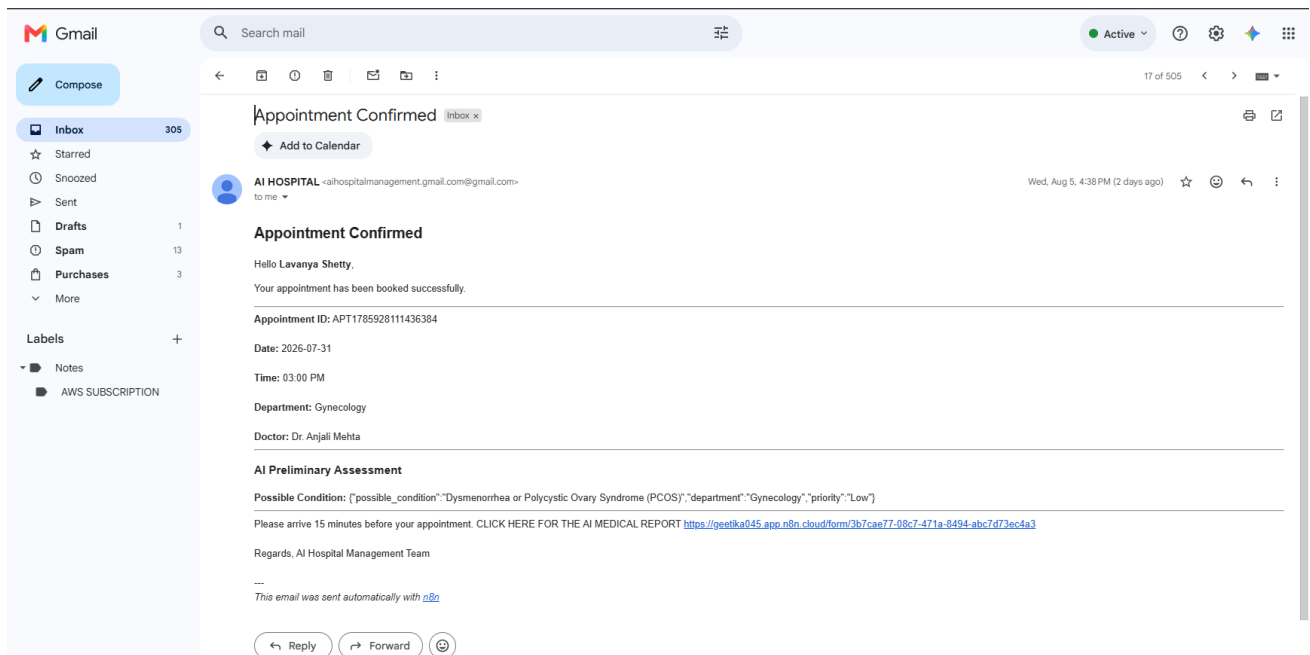
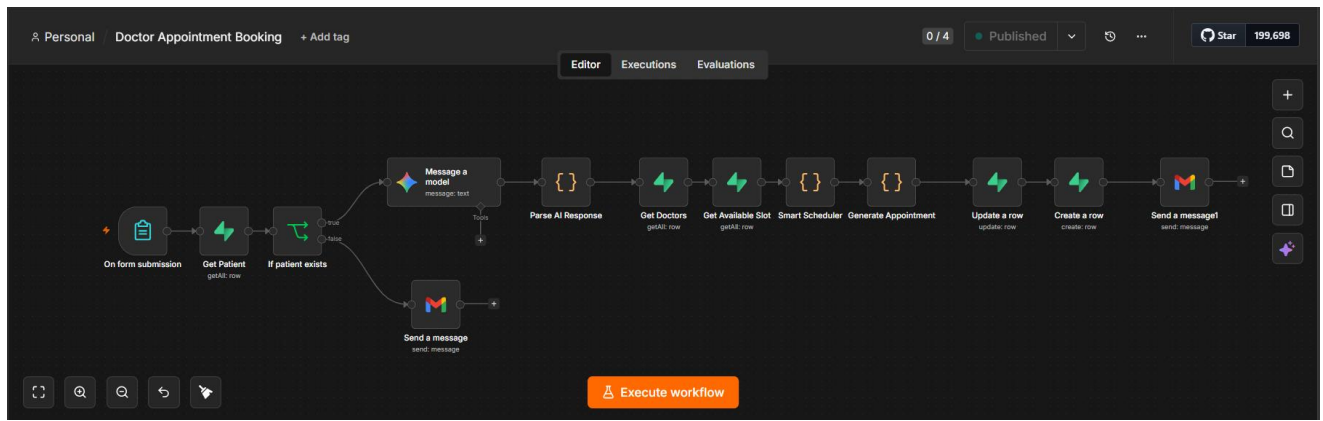
Appointment confirmation email is sent.

Output

- Appointment ID
- Assigned Doctor
- Department
- Confirmation Email

Benefits

- Faster appointment scheduling
- AI-assisted department recommendation
- Reduced manual coordination
- Improved patient experience



6. Workflow 3 – Medical Report Upload & AI Analysis

Objective

This workflow assists doctors by generating a structured preliminary assessment using Google Gemini AI.

Patients upload their medical report, and the workflow extracts text from the document, retrieves patient history, generates AI recommendations, and stores the analysis.

Trigger

Medical report upload form.

Inputs

- Patient ID
- Medical Report (PDF)
- Report Type
- Symptoms

Workflow Process

Step 1

Patient uploads medical report.

↓

Step 2

Text is extracted from the uploaded report.

↓

Step 3

Patient details are retrieved from Supabase.

↓

Step 4

Gemini AI analyzes the available information.

↓

Step 5

AI generates:

- Summary
- Possible Condition
- Severity
- Recommended Department
- Recommended Tests
- Clinical Recommendation

↓

Step 6

Analysis is stored in Supabase.

↓

Step 7

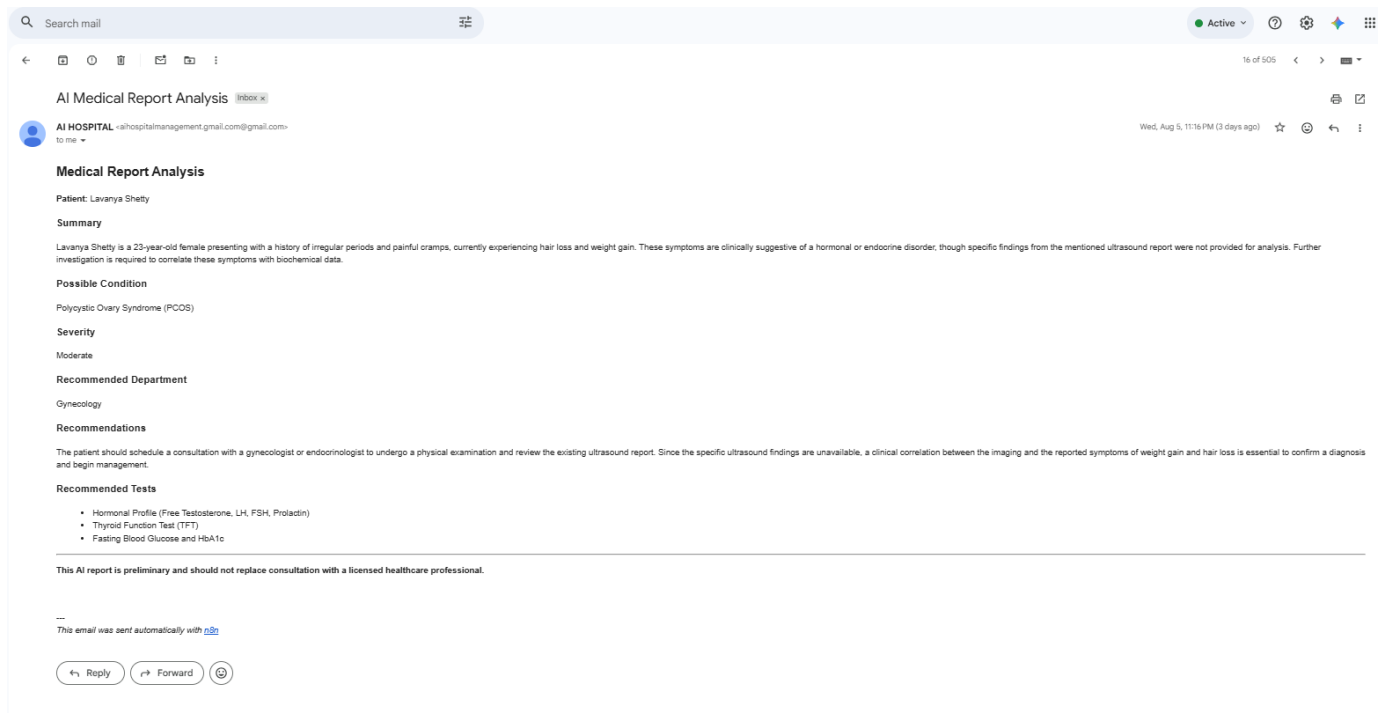
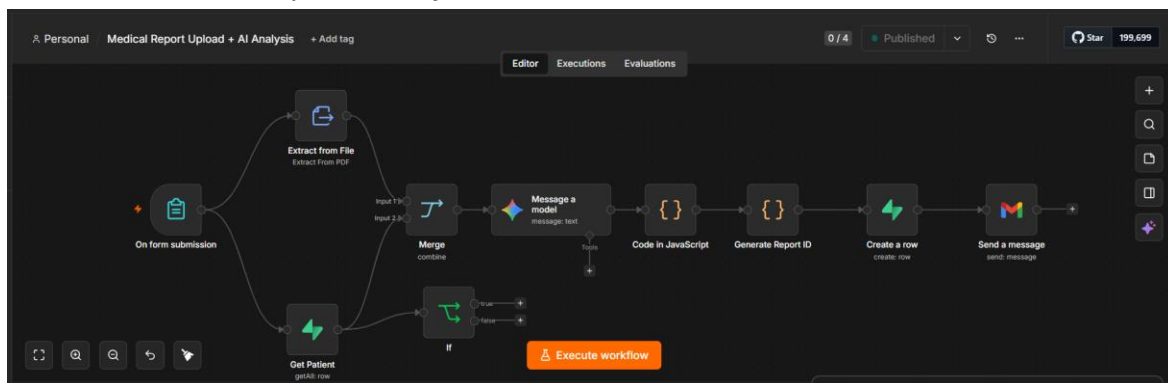
Patient receives notification.

Output

Structured AI Medical Assessment

Benefits

- Supports doctors
- Saves time
- Generates structured reports
- Standardizes preliminary assessment



7. Workflow 4 – AI Doctor Approval System

Objective

Although AI provides recommendations, the final treatment decision should always be made by a qualified doctor.

This workflow enables doctors to review AI-generated assessments and either approve or reject the proposed treatment plan.

Workflow Process

Retrieve Medical Report

↓

Retrieve AI Assessment

↓

Doctor Reviews Information

↓

Approve

OR

Reject

↓

Database Updated

↓

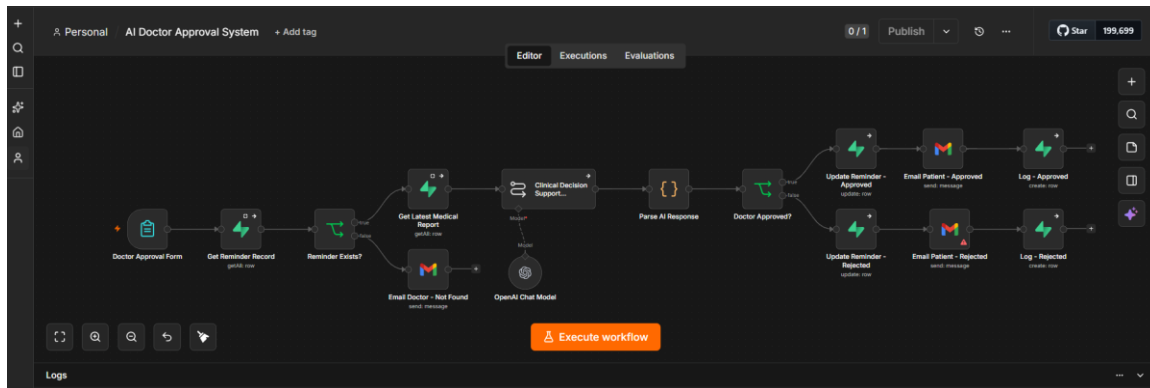
Patient Notified

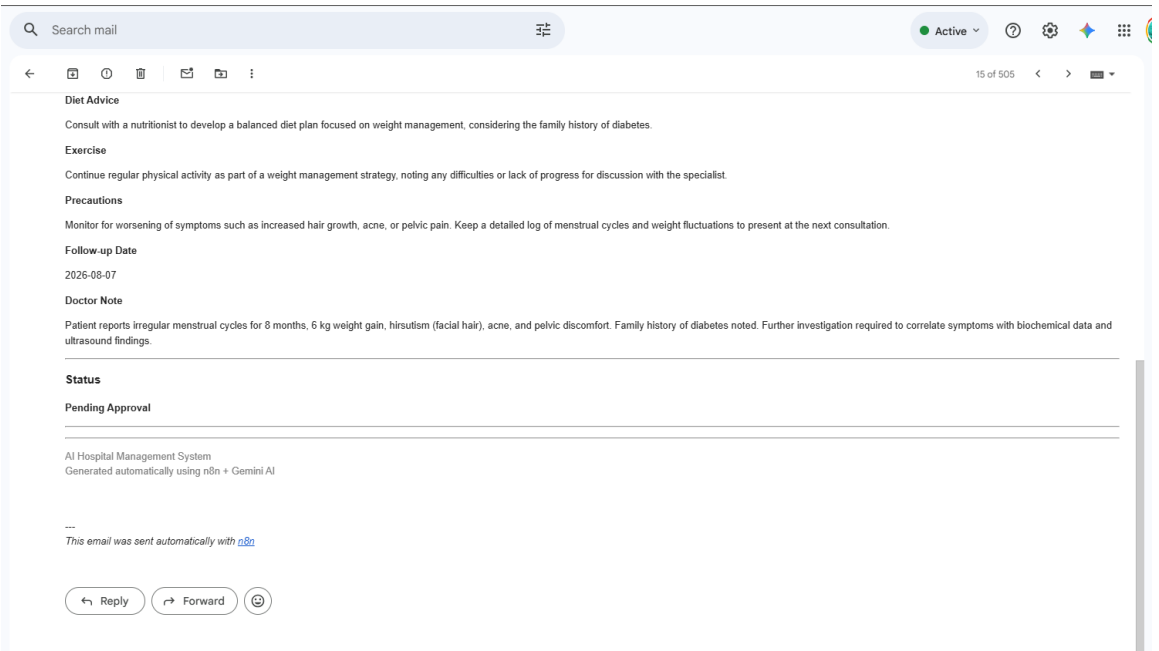
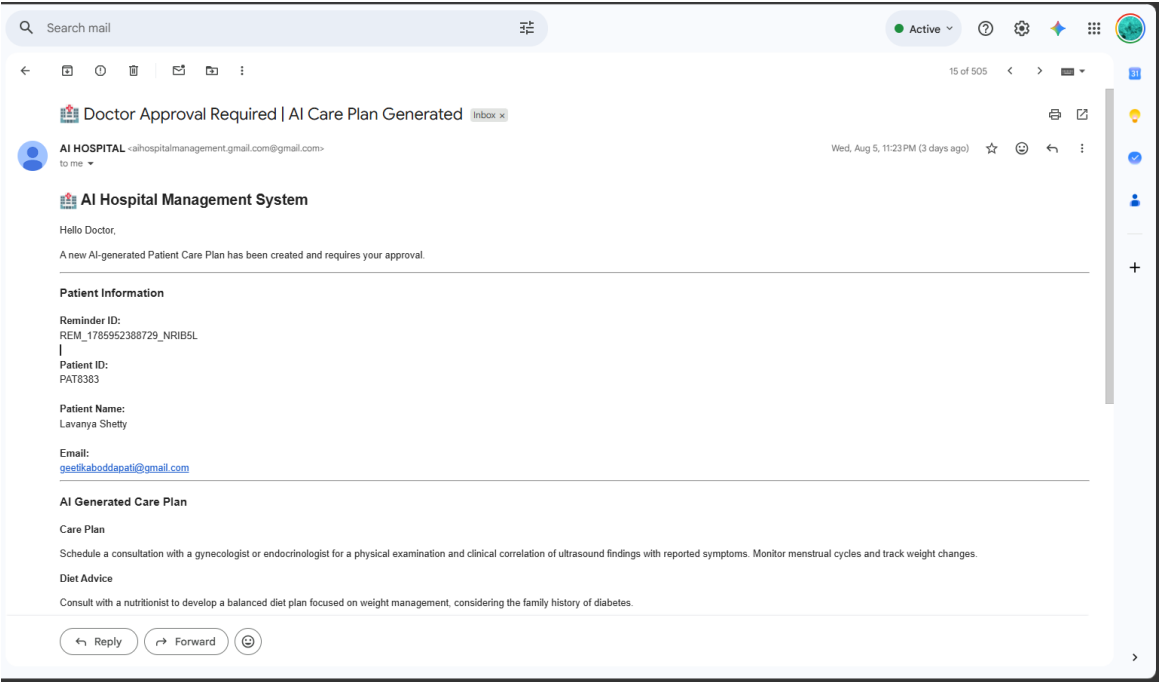
Outputs

- Approval Status
- Doctor Comments
- Updated Medical Record

Benefits

- Human-in-the-loop AI
- Improved patient safety
- Clinical accountability
- Audit trail





8. Workflow 5 – Medication & Follow-up Reminder

Objective

This workflow automatically reminds patients to take medications and attend follow-up consultations.

Workflow Process

Retrieve Reminder Schedule

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Check Reminder Date

↓

Send Reminder Email

↓

Update Reminder Status

↓

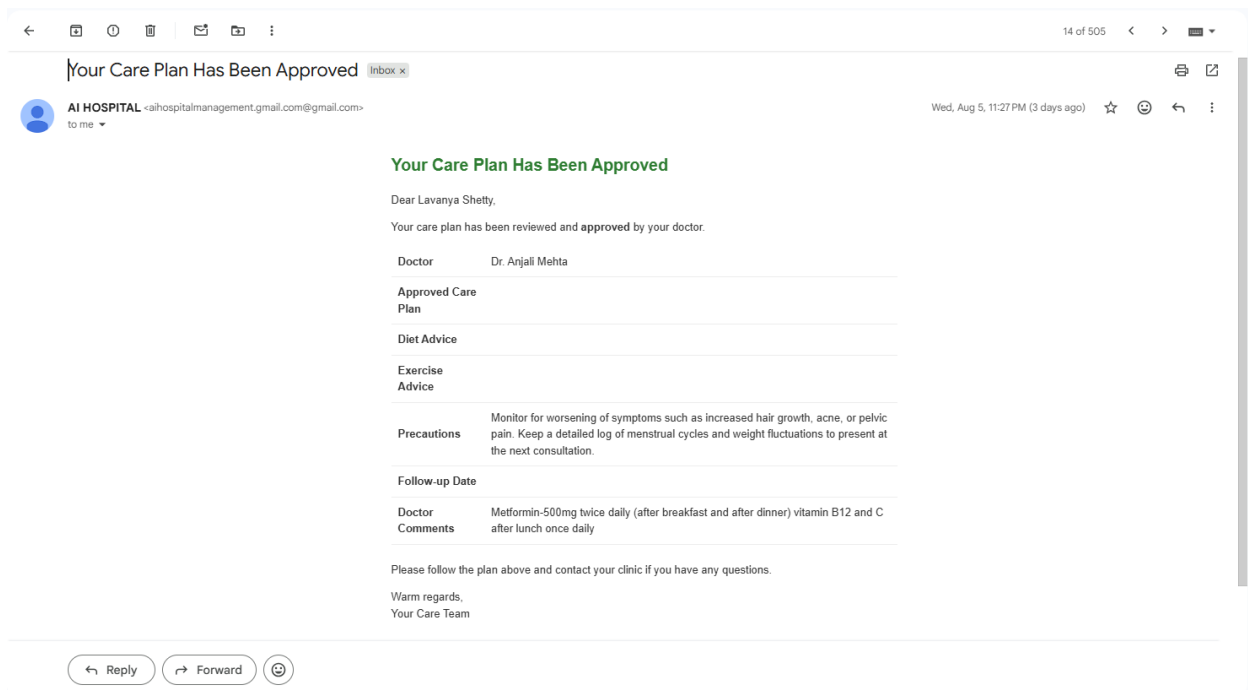
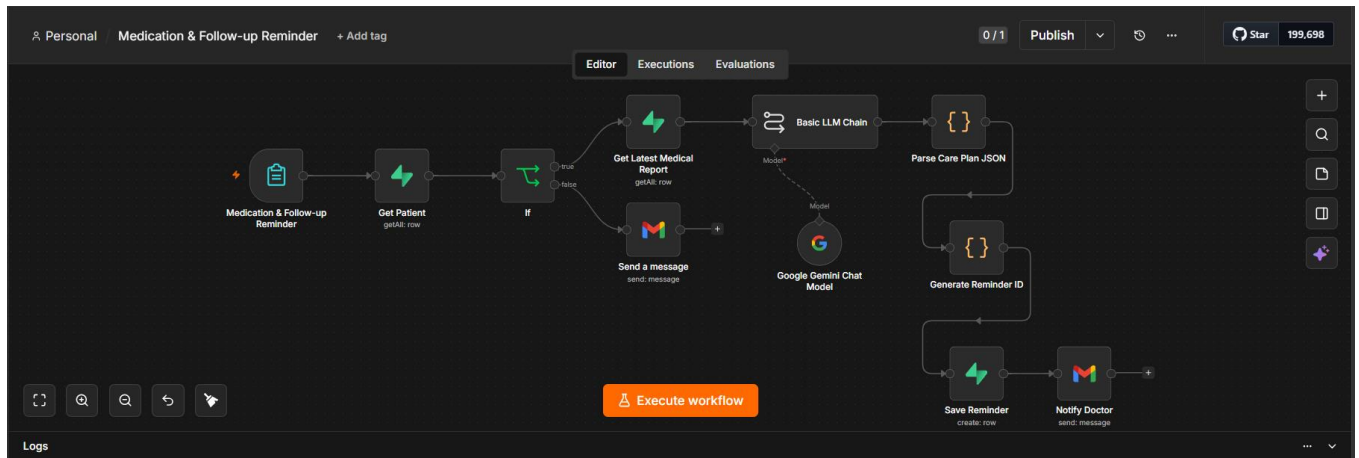
Log Reminder Activity

Outputs

- Reminder Email
- Reminder Status
- Follow-up Schedule

Benefits

- Better medication adherence
- Reduced missed appointments
- Increased patient engagement



9. Workflow 6 – Billing & Hospital Analytics

Objective

The Billing Workflow automates invoice generation after treatment completion.

The workflow calculates consultation fees, medicine costs, laboratory charges, miscellaneous expenses, and taxes before generating the final invoice.

Workflow Process

Retrieve Patient Details

↓

Calculate Charges

↓

Apply GST

↓

Generate Invoice ID

↓

Store Billing Record

↓

Email Invoice

Inputs

- Consultation Fee

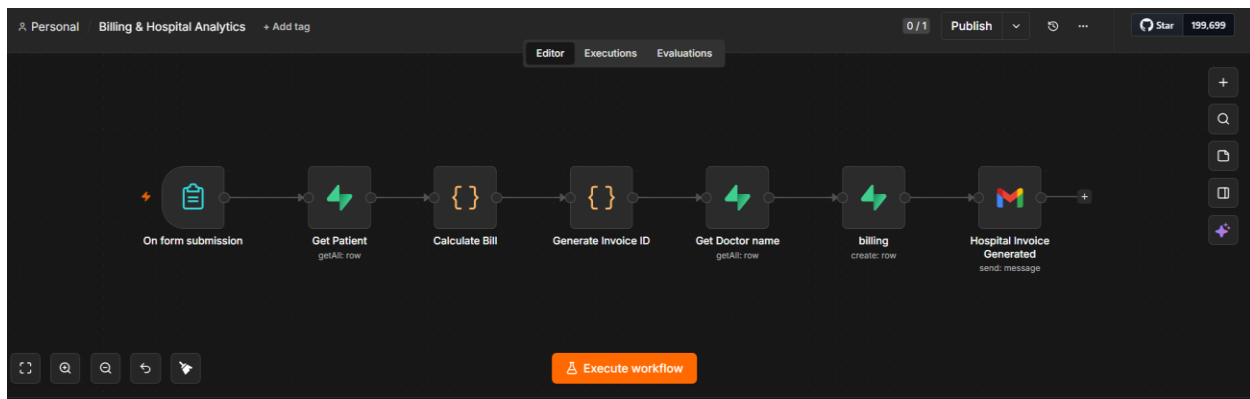
- Medicine Charges
- Laboratory Charges
- Miscellaneous Charges

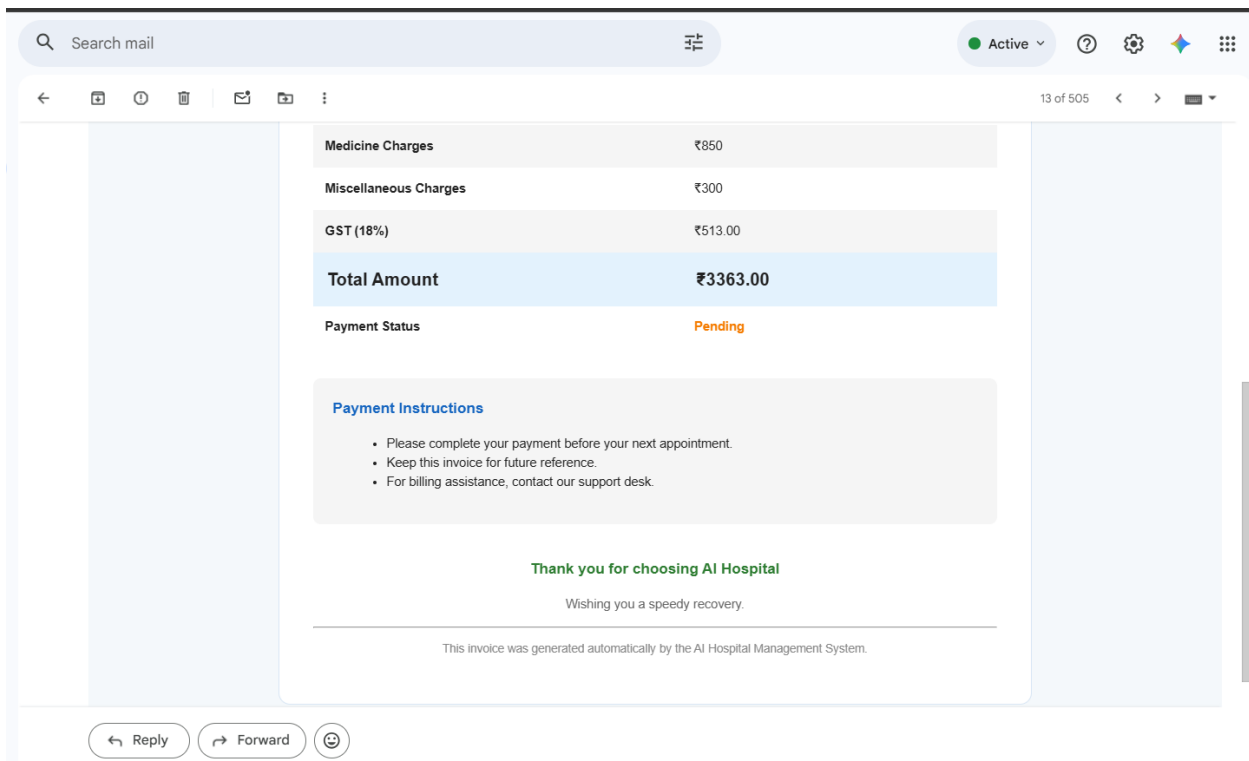
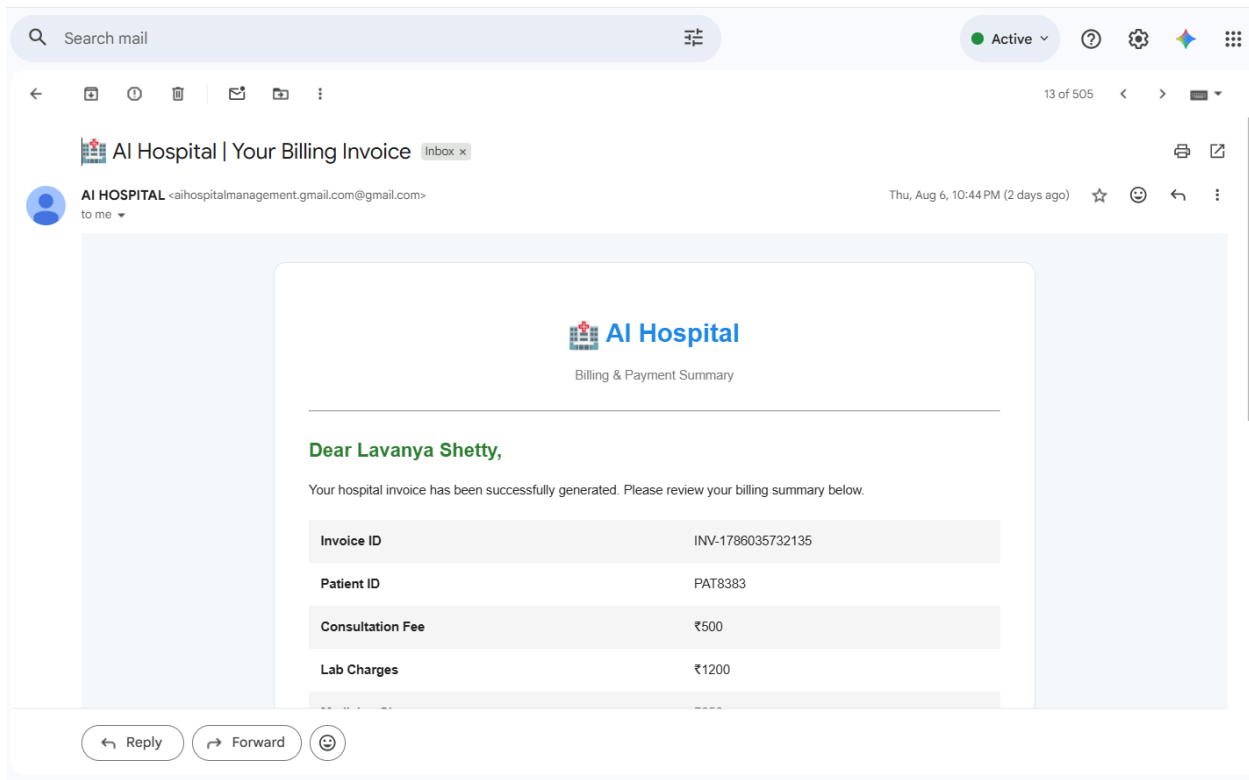
Outputs

- Invoice
- Billing Record
- Email Receipt

Benefits

- Accurate calculations
- Automated invoices
- Faster billing
- Reduced accounting workload





10. Error Handling

The system includes several validation and error-handling mechanisms to ensure reliable execution.

Examples

- Duplicate patient detection
- Invalid Patient ID validation
- Missing doctor availability
- Missing medical report
- AI response validation
- Invalid email detection
- Database insertion failure
- Missing billing information
- Doctor rejection handling

These validations prevent incorrect data from entering the system and improve workflow reliability.

11. Benefits of the System

The AI-Powered Hospital Management System provides numerous operational and clinical benefits.

Operational Benefits

- Faster patient registration
- Automated appointment scheduling
- Centralized patient database
- Automated billing
- Reduced paperwork

Clinical Benefits

- AI-assisted medical assessment
- Human approval before treatment
- Improved patient monitoring
- Better medication compliance

Administrative Benefits

- Automated email notifications
- Reduced manual workload
- Digital audit trail
- Improved workflow efficiency

12. Future Enhancements

The project can be extended with additional capabilities such as:

- Patient Portal
- Doctor Dashboard
- WhatsApp Notifications
- SMS Alerts
- Payment Gateway Integration
- OCR Improvements
- Voice-based Virtual Assistant
- Predictive Disease Analytics
- IoT Medical Device Integration
- FHIR/HL7 Healthcare Standards Support

13. Conclusion

The **AI-Powered Hospital Management System** successfully demonstrates how workflow automation and artificial intelligence can be combined to modernize hospital operations. By integrating **n8n, Google Gemini AI, Supabase, Gmail API, and JavaScript**, the system

automates essential processes such as patient registration, appointment booking, AI-assisted report analysis, doctor approval, medication reminders, and billing.

The project significantly reduces manual effort, improves communication, minimizes processing time, and enhances overall patient care while ensuring that medical decisions remain under the supervision of healthcare professionals. The modular workflow architecture also makes the system scalable, allowing future integration with advanced technologies such as predictive analytics, IoT devices, digital health records, and intelligent patient portals.